EXAM: Topics 1-3 **Bartonian ID _____**

Write neatly ◦ Boxes are reserved for *final answers only* ◦ *Bartonian ID* must be written on each page ◦ Do not staple.

1) State the number of significant figures in each measurement (parts a through c). For part (d), convert the speed to meters per second and report the result with the correct number of significant figures. (1 mi = 1609 m.)

a. 0.0820 m

1a

b. 7.00×10^3 s

1b

c. 150 kg

1c

d. 65.0 mi/h in m/s

1d

2) A hiker walks 4.00 km due east, then 3.00 km in a direction 40.0° north of east. Use the component method to find the hiker's resultant displacement. Give your answers to three significant figures.

a. magnitude (km)

2a

b. direction (° north of east)

2b

3) A car traveling at 24.0 m/s brakes with constant acceleration and comes to rest in a distance of 60.0 m. Take the initial direction of motion as positive. Give your answers to three significant figures.

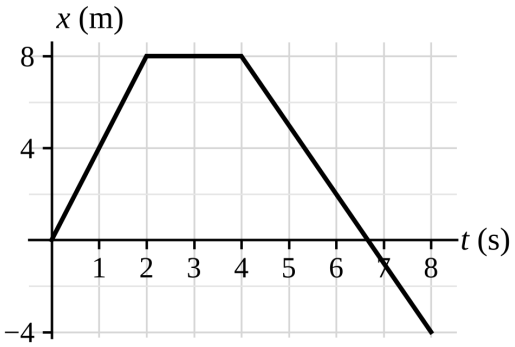
a. acceleration (m/s^2)

3a

b. time to stop (s)

3b

4) The graph below shows position versus time for an object moving along the x-axis. Find the object's average velocity over each interval. (Signs indicate direction.) Give your answers to three significant figures.



a. 0 to 2.00 s (m/s)

4a

b. 0 to 4.00 s (m/s)

4b

c. 4.00 s to 8.00 s (m/s)

4c

d. 0 to 8.00 s (m/s)

4d

5) A ball is thrown straight up at 18.0 m/s. Take upward as positive, ignore air resistance, and use g = 9.80 m/s^2. Give your answers to three significant figures.

a. maximum height above release (m)

5a

b. time to reach the top (s)

5b

c. velocity on return to release level (m/s)

5c

6) A jogger travels 300.0 m due north and then 400.0 m due east, taking a total of 150.0 s. Give your answers to four significant figures.

a. magnitude of displacement (m)

6a

b. direction (° east of north)

6b

c. average speed (m/s)

6c

d. magnitude of average velocity (m/s)

6d

7) A ball rolls off the edge of a table that is 1.20 m above the floor, leaving the edge horizontally at 3.50 m/s. Ignore air resistance and use $g = 9.80\ m/s^2$. Give your answers to three significant figures.

a. time in the air (s)

7a

b. horizontal distance traveled (m)

7b

c. speed just before hitting the floor (m/s)

7c

8) A soccer ball is kicked from level ground with an initial speed of 19.5 m/s at an angle of 30.0° above the horizontal. It lands at the same level from which it was launched. Ignore air resistance and use $g = 9.80\ m/s^2$. Give your answers to three significant figures.

a. maximum height (m)

8a

b. total time of flight (s)

8b

c. horizontal range (m)

8c

9) A river 120 m wide flows due east at 3.00 m/s. A boat that moves at 4.00 m/s relative to the water points its bow due north and crosses from the south bank to the north bank. Give your answers to three significant figures.

a. speed relative to the shore (m/s)

9a

b. direction of travel (° east of north)

9b

c. time to cross (s)

9c

d. distance carried downstream (m)

9d

EQUATION SHEET

Constants and conversions

$g = 9.80\ m/s^2$

1 mi = 1609 m

1 km = 1000 m

1 h = 3600 s

Trigonometry and vectors

$\sin\theta = \text{opp} / \text{hyp}$ $\cos\theta = \text{adj} / \text{hyp}$ $\tan\theta = \text{opp} / \text{adj}$

$A_x = A\cos\theta$ $A_y = A\sin\theta$

$A = \sqrt{(A_x^2 + A_y^2)}$ $\theta = \tan^{-1}(A_y / A_x)$

$R_x = A_x + B_x + \ldots$ $R_y = A_y + B_y + \ldots$

Motion in one dimension (constant a)

$v_{avg} = \Delta x / \Delta t$ average speed = total distance / total time

$a_{avg} = \Delta v / \Delta t$

$v = v_0 + at$

$\Delta x = v_0 t + \frac{1}{2}at^2$

$v^2 = v_0^2 + 2a\Delta x$

$\Delta x = \frac{1}{2}(v_0 + v)t$

Free fall: $a = -g$ (upward positive)

Motion in two dimensions (projectiles)

$v_{0x} = v_0\cos\theta$ $v_{0y} = v_0\sin\theta$

$x = v_{0x}t$ $v_x = v_{0x}$ (constant)

$y = v_{0y}t - \frac{1}{2}gt^2$ $v_y = v_{0y} - gt$ $v_y^2 = v_{0y}^2 - 2g\Delta y$

Level ground: $T = 2v_{0y} / g$ $H = v_{0y}^2 / 2g$ $R = v_0^2 \sin 2\theta / g$

Speed: $v = \sqrt{(v_x^2 + v_y^2)}$

Relative velocity

$v_{AE} = v_{AB} + v_{BE}$ (A relative to E = A relative to B + B relative to E)